

No. of Functional Features Included in the Solution

Date	15 July 2026
Team ID	
Project Name	SkillEcho - Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done/In Progress/Pending)	Marks Contribution
1	User Registration & Login	Account creation and SHA-256 authenticated login with role-based access (student/admin)	main.py (Auth)	Done	High
2	Concept Library (CRUD)	Admin-managed reference concepts with title, text, and optional PDF attachment; 5 seeded concepts	main.py (Concepts)	Done	High
3	Whisper STT Transcription	Converts uploaded/recorded audio into English text using the Whisper tiny model	speech.py	Done	High
4	Topic Guardrail	Vocabulary overlap check that flags off-topic submissions with score 0 before heavy inference	nlp.py	Done	Medium
5	Acoustic Feature & Filler Analysis	Librosa-based RMS energy, zero-crossing rate, pause ratio, plus regex filler-word analysis	audio.py	Done	High
6	Semantic Similarity Scoring	Bi-Encoder (MiniLM-L6-v2) and Cross-Encoder (ms-marco) scores for topic alignment and accuracy	nlp.py	Done	High
7	Weighted Scoring Engine	Combines all metrics into a deterministic 0-100 score with 3-tier classification	scoring.py	Done	High
8	PDF Report Generation	In-memory ReportLab PDF with metrics table, waveform chart, and transcript	reports.py	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	30
Total Features Implemented	30
Core / Must-Have Features	24 (Auth, Concept Mgmt, Audio & Transcription, AI Evaluation Pipeline, Data Persistence)
Additional / Nice-to-Have Features	6 (Admin dashboards, Swagger docs, structured logging extras)

Metric	Count / Value
Features Tested & Verified	30 (100% functional test pass rate)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	6	Audio recorder widget, evaluation scorecard, PDF download button, history view, admin dashboard, concept browser
2	Backend / Logic	7	transcribe_audio(), compute_semantic_similarity(), extract_signal_features(), evaluate_understanding(), build_prompt()-style pipeline orchestration in main.py
3	Database / Storage	2	10-table SQLAlchemy ORM schema, init_db()/seed_db()/get_db() session management
4	API / Integration	12	POST /evaluate, GET /concepts, GET /users/{id}/history, POST /admin/concepts, GET /health, and 7 more REST endpoints
5	Security / Authentication	3	POST /auth/register, POST /auth/login, role-based dashboard routing (student vs. admin)